How do scientists calculate the probability that an asteroid could hit Earth?

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Je me préparais pour ma classe de début de matinée en janvier 2025 quand j'ai reçu une notice concernant un astéroïde appelé 2024 YR4. Il a dit que la probabilité qu'il puisse frapper la Terre était exceptionnellement élevée.

Comme la défense de la Terre contre des intrus inattendus tels que les astéroides fait partie de mon expertise, j'ai immédiatement commencé à recevoir des questions de mes étudiants et de mes collègues sur ce qui se passait.

Lorsque les scientifiques repèrent un astéroide dont la trajectoire pourrait l'emmener près de la Terre, ils le surveillent fréquemment et calculent la probabilité qu'il intervienne en collision avec notre planète. Comme ils reçoivent plus de données d'observation, ils obtiennent une meilleure idée de ce qui pourrait se passer.

Le simple fait d'avoir plus de points de données tôt ne rend pas les prédictions des scientifiques meilleurs. Ils doivent continuer à suivre l'astéroide lorsqu'il se déplace à travers l'espace pour mieux comprendre sa trajectoire.

S'agissant de l'incident quelques mois plus tard, je me suis demandé s'il y avait un meilleur moyen pour les scientifiques de communiquer sur le risque avec le public. Nous avons obtenu des informations précises, mais comme les questions que j'ai entendues l'indiquaient, ce n'était pas toujours suffisant pour comprendre ce que cela signifiait réellement.

Les chiffres changent tous les jours

L'astéroide 2024 YR24 a un diamètre d'environ 60 mètres (60 mètres) - équivalent à environ un bâtiment de 15 étages

en longueur.

At the time of the announcement in January, the asteroid's impact probability was reported to exceed 1%. The impact probability describes how likely a hazardous asteroid is to hit Earth. For example, if the impact probability is 1%, it means that in 1 of 100 cases, it hits Earth. One in 100 is kind of rare, but still too close for comfort if you're talking about the odds of a collision that could devastate Earth.

Over time, though, further observations and analyses revealed an almost-zero chance of this asteroid colliding with Earth.

After the initial notice in January, the impact probability continuously increased up to 3.1% on Feb. 18, but dropped to 1.5% on Feb. 19. Then, the impact probability continuously went down, until it hit 0.004% on Feb. 24. As of June 15, it now has an impact probability of less than 0.0000081%.

But while the probability of hitting Earth went down, the probability of the asteroid hitting the moon started increasing. It went up to 1.7% on Feb. 24. As of April 2, it is 3.8%.

If it hits the moon, some ejected materials from this collision could reach Earth. However, these materials would burn away when they enter Earth's thick atmosphere.

Impact probability

To see whether an approaching object could hit Earth, researchers find out what an asteroid's orbit looks like using a technique called astrometry. This technique can accurately determine an object's orbit, down to only a few kilometers of uncertainty. But astrometry needs accurate observational data taken for a long time.

Any uncertainty in the calculation of the object's orbit causes variations in the predicted solution. Instead of one precise orbit, the calculation usually gives scientists a cloud of its possible orbits. The ellipse enclosing these locations is called an error ellipse.

The impact probability describes how many orbital predictions in this ellipse hit Earth.

Without enough observational data, the orbital uncertainty is high, so the ellipse tends to be large. In a large ellipse, there's a higher chance that the ellipse "accidentally" includes Earth—even if the center is off the planet. So, even if an asteroid ultimately won't hit Earth, its error ellipse might still include the planet before scientists collect enough data to narrow down the uncertainty.

As the level of uncertainty goes down, the ellipse shrinks. So, when Earth is inside a small error ellipse, the impact probability may become higher than when it's inside a large error ellipse. Once the error ellipse shrinks enough that it no longer includes Earth, the impact probability goes down significantly. That's what happened to 2024 YR4.

The impact probability is a single, practical value offering meaningful insight into an impact threat. However, just using the impact probability without any context may not provide meaningful guidelines to the public, as we saw with 2024 YR4.

Holding on and waiting for more data to refine a collision prediction, or introducing new metrics for assessing impacts on Earth, are alternative courses of action to provide people with better guidelines for future threats before adding confusion and fear.

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